

Oral Exam Syllabus

Avinash Iyer

October 2026

The oral exam for Avinash Iyer will be administered by Professors Ben Hayes and Mikhail Ershov on December 4, 2026 focused on Functional Analysis and Geometric Group Theory. The exam will consist of questions on both topics.

Functional Analysis Topics:

- Locally Convex Spaces, topics covering Chapter IV §1–§3 of Conway's *A Course in Functional Analysis*
 - Definitions and examples
 - Metrizability and Normability
 - Geometric Hahn–Banach
- Weak topologies and extremal structure, topics covering Chapter V §1–§10 §12–§13 of Conway's *A Course in Functional Analysis*
 - Dual spaces of locally convex topologies
 - Weak compactness, Banach–Alaoglu Theorem
 - Reflexivity, Separability, and Metrizability
 - Krein–Milman theorem
- C^* -algebras and von Neumann algebras, topics covering Chapter VIII §1–§3, §5 of Conway's *A Course in Functional Analysis* and Sections 4.6 and 4.7 of Pedersen's *Analysis Now*
 - Properties and examples
 - Gelfand Transform, Gelfand–Naimark Theorem, Continuous Functional Calculus
 - Ordering properties
 - GNS Construction
 - von Neumann double commutant theorem
 - σ -topologies and the predual of a von Neumann algebra
- Spectral theory for normal operators, topics covering Chapter IX §1–§8 of Conway's *A Course in Functional Analysis* and Sections 4.4–4.6 of Pedersen's *Analysis Now*
 - spectral measures and projections
 - Borel functional calculus
 - the characterization of abelian von Neumann algebras
- Comparison theory of projections in a von Neumann Algebra (various sources):
 - Equivalence and sub-equivalence of projections;

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- Schröder–Bernstein theorem for projections;
 - Comparison Theorem;
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Geometric Group Theory:

- Relations between groups and their action on graphs, topics covering Chapters 3 and 4 of Löh’s *Geometric Group Theory*
 - Cayley graphs
 - actions of groups on trees; proof of the Nielsen–Schreier Theorem
- Quasi-isometry, topics covering Chapter 5 and 6 of Löh’s *Geometric Group Theory* and sections 11.12–11.13 of Ceccherini-Silberstein and d’Adderio’s *Topics in Groups and Geometry*
 - definition of quasi-isometry of metric spaces and groups, quasi-geodesics;
 - Švarc–Milnor lemma and its topological formulation (statement and applications)
 - dynamic criterion for quasi-isometry (set-theoretic coupling and topological coupling)
 - asymptotic cones
 - notion of growth rate
 - quasi-isometry invariance of growth rate
- Amenability, topics covering Chapter 9 of Löh’s *Geometric Group Theory* and other sources
 - definition via means and Følner Sequences, elementary inheritance
 - proofs of sufficient criteria (polynomial growth, solvability)
 - equivalent criteria (paradoxicality, almost-invariant vectors, weak containment)
 - quasi-isometry invariance of amenability
- Soficity, topics covering content from Pestov’s *Hyperlinear and Sofic Groups*, Pestov and Kwiatowska’s *An Introduction to Hyperlinear and Sofic Groups*, and Ceccherini-Silberstein and Coornaert’s *Cellular Automata and Groups*
 - almost-homomorphism into symmetric groups;
 - embedding into metric ultraproduct (and sofic approximations);
 - initial subamenability of Cayley graph;
 - inheritance properties and prominent examples (amenable groups, residually finite groups)